Downstream Utility of Synthetic Data for Clustering

Oriol Farrés Vilar[1], Jordi Cortés[2], Xinnuo Chen[3], Victoria López[4], Daniel Fernández[2]

[1]Department of Computer Science, Universitat Politècnica de Catalunya, Barcelona, Spain.
[2]Department of Statistics and Operations Research, Universitat Politècnica de Catalunya, Barcelona, Spain.
[3]Department of Statistical Science, University College London, London, United Kingdom.
[4]CUNEF Universidad, Madrid, Spain.

Abstract

Separate title page of a manuscript submitted to the *Journal of Classification*; the manuscript itself is anonymised for double-anonymous review. [**TO COMPLETE:** paste the final abstract ($\leq$ 150 words) here]

Keywords: Synthetic data, Cluster analysis, Statistical disclosure control, Analytical utility, Simulation study

Corresponding author

[**TO COMPLETE:** name of the corresponding author]
E-mail: [**TO COMPLETE:** institutional e-mail address]
Postal address: [**TO COMPLETE:** full postal address]
Telephone: [**TO COMPLETE:** telephone number, with country code]
(The JoC submission page asks for e-mail, full postal address and telephone number of the submitting author.)

ORCID identifiers

Oriol Farrés Vilar: [**TO COMPLETE:** ORCID]
Jordi Cortés: [**TO COMPLETE:** ORCID]
Xinnuo Chen: [**TO COMPLETE:** ORCID]
Victoria López: [**TO COMPLETE:** ORCID]
Daniel Fernández: [**TO COMPLETE:** ORCID]

Acknowledgements

[**TO COMPLETE:** acknowledgements, or delete this section]

Declarations

Funding. [**TO COMPLETE:** grant agencies and grant numbers for each author, or "The authors did not receive support from any organization for the submitted work."]

Competing interests. [**TO COMPLETE:** "The authors have no relevant financial or non-financial interests to disclose." or the relevant disclosure; one statement covering all five authors]

Author contributions. [**TO COMPLETE:** contribution of each author, e.g. in CRediT terms (Conceptualization, Methodology, Software, Formal analysis, Writing – original draft, Writing – review & editing, Supervision). To be agreed by all five authors.]

Data availability. All data analysed in this study are simulated; they can be regenerated exactly from the configuration file and seeds distributed with the code. [**TO COMPLETE:** add a DOI (e.g. Zenodo/figshare) if the simulated datasets are also deposited]

Code availability. The full pipeline, from data generation through to the figures and tables reported in the article, is available at https://github.com/Sterrysx/synthetic-clustering. [**TO COMPLETE:** archived release with DOI, and the anonymised mirror used for review, e.g. https://anonymous.4open.science/r/XXXX]

Ethics approval and consent. Not applicable: the study uses only simulated data.